

Samples + specification

Source and conditions
Units and requirements

Clearance ≥ 5 m

32 calibration samples

Calibration + training

Versioned speed envelope



Loading context

E_0 : simulation

Nominal / extended scope



Stress tests and haul cycles

Evidence ledger

Trace, verdict, model hash



Scenario-linked replay

Human verifier

Review failure cases

Assess uncertainty

Approve physical tests

Integrated revision

Controller and planner
Deployment orchestrator



4,000 validation tests

E_1 : controlled ground

Braking tests

Detection tests

Physical evidence
pending

E_2 : supervised field

Site-specific testing

Physical evidence
pending

Release blocked

Physical confirmation and signed approval are missing

—→ Implemented simulation workflow
- - - → Proposed physical confirmation